Kagami-Ogata Syndrome (KOS14): A Comprehensive Disease Characterization Report

MONDO ID: MONDO:0016779 · **OMIM:** #608149 · **Orphanet:** ORPHA:97570 (UPD(14)pat) · **Category:** Mendelian / genomic-imprinting disorder

Summary

Kagami-Ogata syndrome (KOS) is not caused by a classical single-gene mutation but by an imbalance of genomic imprinting at the chromosome 14q32.2 DLK1-MEG3 (DLK1-DIO3) imprinted cluster. The disease results from a **net excess of paternally expressed genes — chiefly the retrotransposon-derived gene RTL1/PEG11 and DLK1 — with concomitant loss of maternally expressed non-coding RNAs** (MEG3/GTL2, the RTL1 antisense transcript RTL1as, MEG8, and clustered miRNAs/snoRNAs). There is therefore no single "causal gene" in the Mendelian sense; rather, the disease is defined by the dosage of an imprinted gene cluster. Among the imprinted genes, **RTL1 overdosage is the principal driver** of the characteristic phenotype, with DLK1 contributing to the growth and metabolic features.

Three distinct molecular mechanisms converge on the same lesion — a "double dose paternal / zero dose maternal" state at 14q32.2: **paternal uniparental disomy of chromosome 14 (upd(14)pat, ~65%), maternally-derived microdeletions (~19%), and epimutations (hypermethylation of the IG-DMR/MEG3-DMR, ~15%)**. The clinical result is a pathognomonic gestalt: a small, bell-shaped thorax with "coat-hanger" angled ribs, a distinctive face (full cheeks, protruding philtrum, depressed nasal bridge), abdominal wall defects, polyhydramnios, placentomegaly, and birth overgrowth. Postnatally, severe respiratory insufficiency from the constricted thorax dominates the course and drives a poor prognosis; survivors have universal developmental delay (median IQ/DQ ~55) and an elevated (~8.8%) risk of hepatoblastoma.

KOS is the **molecular mirror image of Temple syndrome (TS14)**, which arises from the reciprocal dosage state (maternal UPD / paternal deletion or hypomethylation). The dosage mechanism is validated across species by mouse models (Rtl1 and miR-127 knockouts) and by the naturally occurring ovine **callipyge** muscular-hypertrophy phenotype, both of which demonstrate that gene dosage at this cluster — not coding mutation — is the pathogenic variable. No disease-modifying therapy exists; diagnosis relies on recognizing the radiographic gestalt confirmed by methylation/UPD/CNV testing, and management is supportive.

Key Findings

Finding 1 — KOS is caused by excess paternal / absent maternal gene dosage at the 14q32.2 DLK1-MEG3 imprinted locus

The foundational finding of this investigation is that **KOS is a genomic-imprinting disorder rather than a conventional coding-variant disease**. Three molecular mechanisms all produce the same functional endpoint — overexpression of paternally-expressed genes and silence of maternally-expressed non-coding RNAs at 14q32.2:

1. **Paternal uniparental disomy 14 (upd(14)pat)** — two paternal copies, no maternal copy.
2. **Maternal-origin microdeletions** — loss of the maternal allele/DMR.
3. **Epimutations** — aberrant hypermethylation of the maternally-inherited IG-DMR (MEG3/DLK1:IG-DMR) and MEG3:TSS-DMR, silencing the maternal program.

In a cohort of 26 patients with the UPD(14)pat-like phenotype, the relative frequencies were **upd(14)pat 65.4% (17/26), microdeletions 19.2% (5/26), and epimutations 15.4% (4/26)** (PMID: 22353941): *"we identified UPD(14)pat in 17 patients (65.4%), microdeletions of different patterns in 5 patients (19.2%), and epimutations in 4 patients (15.4%)."*

The mechanistic through-line is **RTL1 overexpression**. When the maternally expressed RTL1as (antisense) transcript — which encodes microRNAs that repress RTL1 — is lost, RTL1 is de-repressed. RTL1 placental expression was **~5× control** in patient placentas (PMID: 22917972): *"RTL1 expression level was about five times higher in the placental samples of the two patients than in control placental samples"*, and ~2.5× in body tissues. A review concluded that *"excessive RTL1 expression and absent MEG expression constitute the primary underlying fact"* (PMID: 26377239).

Molecular mechanism	Frequency	Net effect at 14q32.2
Paternal UPD14 (upd(14)pat)	~65%	2× paternal, 0× maternal
Maternal microdeletion	~19%	Loss of maternal DMR/genes
Epimutation (IG-DMR/MEG3-DMR hypermethylation)	~15%	Maternal allele epigenetically silenced

Finding 2 — KOS has a pathognomonic clinical gestalt with developmental delay and ~8.8% hepatoblastoma risk

In the landmark cohort of **34 molecularly-defined patients** (UPD(14)pat n=23, epimutations n=5, microdeletions n=6), the defining clinical features were a **distinctive face** (full cheeks, protruding philtrum) and an **increased coat-hanger rib angle** on chest X-ray with a small bell-shaped thorax (PMID: 25689926). These two features together are considered pathognomonic and sufficient to prompt molecular confirmation.

Quantitatively: - **Birth overgrowth:** median birth weight **+2.3 SD** (range +0.1 to +8.8 SD) — *"a median birth weight of +2.3 SD (range, +0.1 to +8.8 SD)"*. - **Developmental delay:** *"developmental delay and/or intellectual disability was invariably present, with a median developmental/intellectual quotient of 55 (range, 29–70)"* — i.e., 100% penetrant with moderate severity. - **Hepatoblastoma:** *"hepatoblastoma was identified in three infantile patients (8.8%)"* (3/34), paralleling the tumor predisposition of another imprinting disorder, Beckwith-Wiedemann syndrome.

Prenatally the syndrome presents with **polyhydramnios and placentomegaly** (PMID: 25689926, PMID: 33179779). The postnatal course is dominated by **severe respiratory distress** from the restrictive small thorax (PMID: 33179779, PMID: 34746047), which is the principal determinant of early mortality.

Finding 3 — Mouse models confirm Rtl1 dosage as the driver; both overexpression and loss are lethal

The dosage model is directly supported by mouse genetics. **Rtl1 (Peg11)**, a retrotransposon-derived paternally expressed gene on mouse chromosome 12 (syntenic to human 14q32), is essential for maintenance of fetal capillaries at the feto-maternal interface. Critically, **both its**

loss and its overproduction cause late-fetal/neonatal lethality (PMID: 18176565): *"both its loss and its overproduction cause late-fetal and/or neonatal lethality in mice"* — the clearest demonstration that this gene is exquisitely dosage-sensitive, recapitulating KOS lethality.

The repressor arm is equally informative: **miR-127**, processed from the maternally expressed Rtl1as antisense transcript, is an essential trans-acting repressor of Rtl1. **miR-127 knockout mice exhibit placentomegaly** with labyrinthine-zone defects and increased Rtl1 transcript/protein (PMID: 26138477): *"The miR-127 knockout mice exhibit placentomegaly with specific defects within the labyrinthine zone involved in maternal-fetal nutrient transfer"* — directly linking loss of the maternal microRNA repressor to the placentomegaly hallmark of KOS. Mouse chromosome 12qF1 is strongly syntenic to human 14q32 (PMID: 32592473).

Finding 4 — Diagnosis rests on the radiographic gestalt confirmed by methylation/UPD/CNV testing; management is supportive

Prenatal clues that should trigger investigation are **severe polyhydramnios + small bell-shaped thorax + protuberant abdomen + dysmorphic face** (PMID: 33179779). Confirmatory molecular testing uses a tiered approach: - **Methylation-specific MLPA (MS-MLPA)** of the IG-DMR/MEG3-DMR and methylation-specific PCR; - **SNP microarray** and microsatellite/UPD analysis (to detect upd(14)pat); - **Karyotype/FISH** to detect Robertsonian translocations predisposing to UPD; - **CNV-seq** for microdeletions.

"Prenatal diagnosis using methylation-specific multiplex ligation-dependent probe amplification (MLPA) or a single-nucleotide polymorphism-based microarray analysis is recommended" (PMID: 34746047; see also PMID: 36035167, PMID: 32211354).

There is **no disease-modifying therapy**. Management is supportive: mechanical ventilation/tracheostomy for respiratory insufficiency, surgical repair of abdominal wall defects (omphalocele/diastasis recti), feeding support, developmental/physical therapy, and **periodic hepatoblastoma surveillance (serum AFP + abdominal ultrasound)** (PMID: 25689926, PMID: 31853915): *"These findings suggest the necessity of an adequate support for developmental delay and periodical screening for hepatoblastoma in the affected patients."*

Finding 5 — KOS is the molecular mirror image of Temple syndrome; DLK1 dosage links the locus to growth, puberty, and metabolism

KOS and **Temple syndrome (TS14)** are reciprocal imprinting disorders of the identical cluster (PMID: 32592473): *"Temple syndrome (TS) and Kagami-Ogata syndrome (KOS) are imprinting disorders caused by absence or overexpression of genes within a single imprinted cluster on*

human chromosome 14q32." TS14 arises from maternal UPD14 / paternal deletion or hypomethylation (the *opposite* dosage state), and presents with growth restriction, hypotonia, small hands/feet, and precocious puberty.

The locus encodes **paternally expressed DLK1, RTL1, DIO3** and **maternally expressed lncRNAs MEG3/GTL2, RTL1as, MEG8** plus clustered miRNAs/snoRNAs. **DLK1 is dosage-sensitive**: loss-of-function causes central precocious puberty with obesity and metabolic derangement (PMID: 36577869, PMID: 31460623): *"DLK1 is an imprinted gene, whose mutations have been described as a rare cause of CPP in girls and adult women with precocious menarche, obesity and metabolic derangement."* By inference, the DLK1 *over*expression in KOS contributes to its growth/metabolic phenotype — the dosage-opposite of the DLK1-loss phenotype.

Finding 6 — The IG-DMR is the master imprinting control element; KOS placental pathology localizes to villous endothelial cells and pericytes

The **IG-DMR is the master maternal-allele imprinting control center**. In mouse, deletion of the IG-DMR from the maternally inherited chromosome causes **bidirectional loss of imprinting of all genes in the cluster**, whereas paternal transmission of the same deletion leaves imprinting unchanged (PMID: 12937418): *"deletion of the IG-DMR from the maternally inherited chromosome causes bidirectional loss of imprinting of all genes in the cluster."* This explains why epimutations of the maternal IG-DMR in KOS recapitulate the full molecular phenotype.

KOS placental pathology has a specific **cellular localization**: terminal villi show swollen vascular endothelial cells and hypertrophic pericytes with narrowed capillary lumens, and DLK1, RTL1, and DIO3 proteins localize specifically to vascular endothelial cells and pericytes, with staining intensity tracking gene dosage (PMID: 22917972): *"Terminal villi were associated with swollen vascular endothelial cells and hypertrophic pericytes, together with narrowed capillary lumens."* Mouse uniparental-disomy studies link Dlk1/Gtl2 dosage to skeletal muscle maturation, bone formation, and placental defects (PMID: 17449025). Notably, **epimutation cases can be milder**, sometimes with near-normal development, reflecting mosaic methylation (PMID: 35953028).

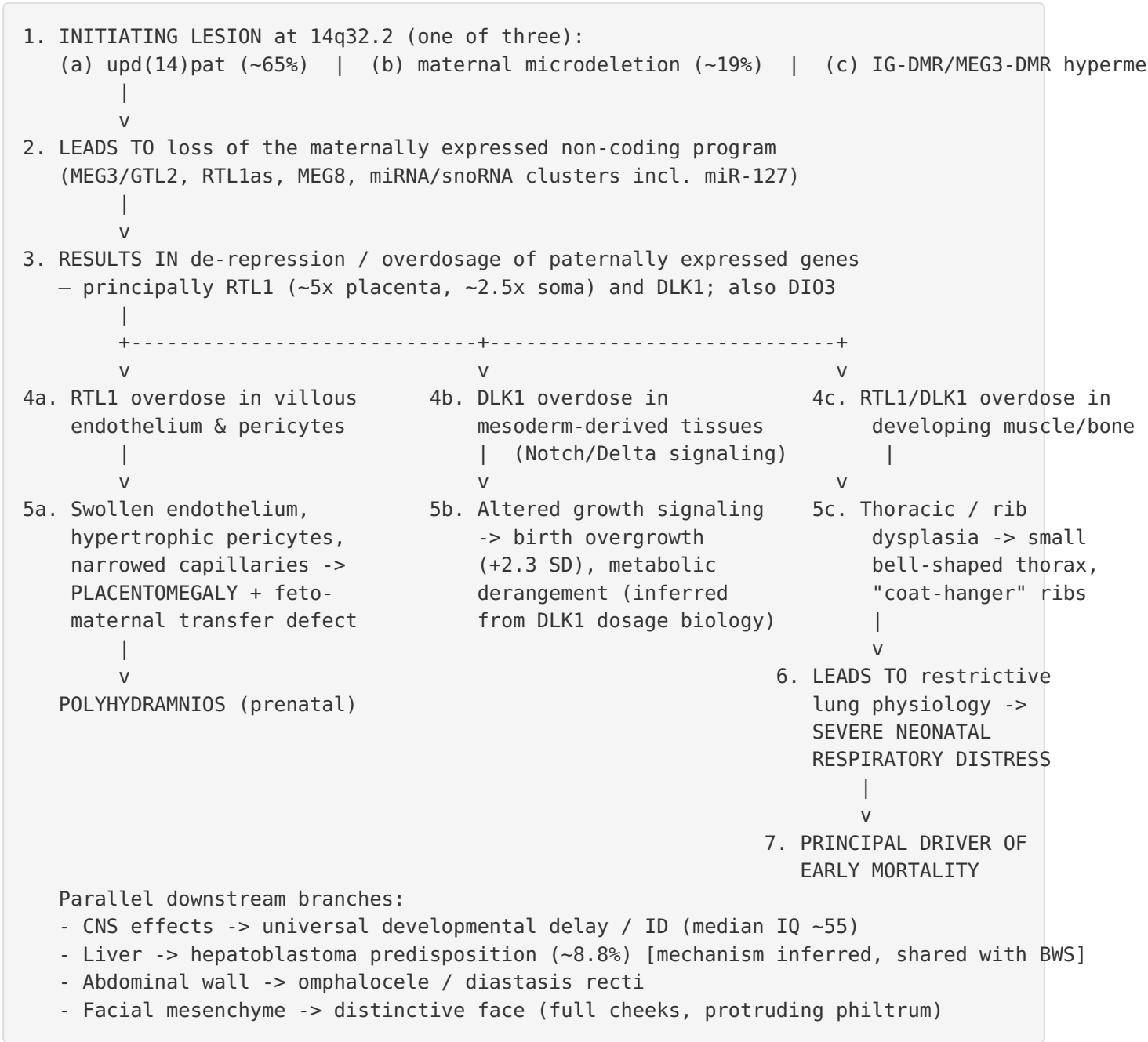
Finding 7 — The ovine callipyge phenotype is a naturally occurring DLK1-DIO3 imprinting disorder that validates the dosage mechanism

In sheep (*Ovis aries*), the **callipyge (CLPG)** phenotype — postnatal muscular hypertrophy — is caused by a single **A→G SNP in a muscle-specific long-range control element** within the DLK1-GTL2 intergenic (IG) region. The mutation up-regulates paternally expressed DLK1/RTL1 and maternal ncRNAs **in cis without altering imprinting status** (PMID: 24990181, PMID: 16690740): *"A single nucleotide polymorphism in the DLK1-DIO3 imprinted gene cluster alters gene expression of the paternal allele-specific protein-coding genes and several maternal allele-specific long noncoding RNA and microRNA when the mutation is inherited in cis."*

Callipyge displays **non-Mendelian polar overdominance** (hypertrophy only in +mat/CLPGpat heterozygotes), explained by trans-inhibition of DLK1 by maternally expressed miRNAs (miR-379/miR-544 cluster, including miR-329) (PMID: 26487685, PMID: 20944086). Crucially, **DLK1 and RTL1 — not MEG3/MEG8 — drive the muscle gene-expression changes** (PMID: 19816583): *"The muscle-specific expression patterns of most genes were similar to DLK1 and included genes that are transcriptional repressors or affect feedback mechanisms in beta-adrenergic and growth factor signaling pathways."* This natural model independently confirms that DLK1/RTL1 overdosage drives musculoskeletal reprogramming, the same mechanistic axis implicated in KOS.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)



Upstream vs downstream. The upstream, proximal event is always a change in *dosage/imprinting state* at 14q32.2 (not a coding mutation). The master control point is the **IG-DMR**, whose maternal methylation state governs the entire cluster (PMID: 12937418). Immediately

downstream is the loss of maternal ncRNAs (especially the RTL1as-derived miR-127) and consequent **RTL1/DLK1 protein overdosage**. The most distal manifestations — respiratory failure, developmental delay, hepatoblastoma — are tissue-specific consequences of this overdosage.

Cell types and processes. Documented cellular targets are **villous vascular endothelial cells (CL:0000115)** and **pericytes (CL:0000669)** in the placenta (PMID: 22917972), with **skeletal muscle (CL:0000188)** and bone-forming cells implicated by cross-species and UPD-mouse data. Relevant biological processes include **genomic imprinting (GO:0071514)**, **regulation of gene expression by genomic imprinting (GO:0006349)**, **Notch signaling pathway (GO:0007219)** (DLK1 is a Notch/Delta-family modulator), **placenta development (GO:0001890)**, and **negative regulation of gene expression by miRNA (GO:0035195)**.

Key chemical/molecular entities. 5-methylcytosine (CHEBI:27551) at the DMRs is the central epigenetic mark; serum **alpha-fetoprotein** is the hepatoblastoma surveillance marker.

Cross-species validation table

System	Lesion	Phenotype	Mechanistic lesson	PMID
Human KOS	upd(14)pat / del / epimutation	Coat-hanger ribs, respiratory failure, DD, hepatoblastoma	Paternal-excess dosage is pathogenic	25689926
Human TS14	mat UPD / pat del / hypomethylation	Growth restriction, precocious puberty	Reciprocal/mirror dosage	32592473
Mouse Rtl1	Loss **or** overexpression	Fetal/neonatal lethality	Rtl1 is dosage-sensitive	18176565
Mouse miR-127 KO	Loss of maternal repressor	Placentomegaly, ↑ Rtl1	Maternal miRNA restrains Rtl1	26138477
Sheep callipyge	cis IG SNP	Muscle hypertrophy (polar overdominance)	DLK1/RTL1 overdosage reprograms muscle	24990181, 19816583

Section-by-Section Report

1. Disease Information

KOS14 is a rare congenital genomic-imprinting disorder of chromosome 14q32.2, first described clinically by Wang et al. in 1991. **Identifiers:** OMIM #608149; MONDO:0016779; Orphanet ORPHA:97570 ("Paternal uniparental disomy of chromosome 14"). **Synonyms:** Kagami-Ogata syndrome; UPD(14)pat syndrome; paternal uniparental disomy 14 syndrome; "coat-hanger rib" syndrome. The knowledge base here is **aggregated disease-level** evidence drawn from published molecularly-defined cohorts (the 34-patient Kagami-Ogata cohort, PMID: 25689926) and case series/reports (PMID: 34055463), not individual EHR data.

2. Etiology

Primary cause: genetic/epigenetic imbalance (not environmental or infectious). Three mechanisms — upd(14)pat (~65%), maternal microdeletion (~19%), epimutation (~15%) (PMID: 22353941). **Genetic risk factor:** Robertsonian translocations (e.g., ROB 13;14) predispose to upd(14)pat through trisomy/monosomy rescue (PMID: 32211354). **Environmental/lifestyle/infectious factors:** none established. **Protective factors:** none known; however, **mosaic/partial epimutation** attenuates severity (PMID: 35953028). **Gene-environment interactions:** none documented; this is a predominantly genetic/epigenetic disorder.

3. Phenotypes

Phenotype	HPO suggestion	Type	Onset	Frequency / severity
Bell-shaped/small thorax	HP:0001591 / HP:0005257	Physical/ skeletal	Prenatal–neonatal	Near-universal; pathognomonic
Coat-hanger rib configuration	HP:0000889	Radiographic sign	Neonatal	Pathognomonic
Neonatal respiratory distress	HP:0002643	Clinical sign	Neonatal	Very frequent; severe
Polyhydramnios	HP:0001561	Prenatal sign	Prenatal	Frequent
Placentomegaly	HP:0100767	Physical	Prenatal	Frequent
Birth overgrowth / high birth weight	HP:0001520	Physical	Congenital	Median +2.3 SD
Developmental delay / intellectual disability	HP:0001263 / HP:0001249	Behavioral/ cognitive	Childhood	**100%**, median IQ ~55
Distinctive face (full cheeks, protruding philtrum, depressed nasal bridge)	HP:0000271	Physical	Congenital	Frequent
Abdominal wall defect (omphalocele, diastasis recti)	HP:0001537	Physical	Congenital	Frequent
Muscular hypotonia	HP:0001252	Clinical sign	Neonatal	Frequent
Hepatoblastoma	HP:0002884	Neoplasm	Infancy	~8.8%

Evidence: PMID: 25689926, PMID: 33179779, PMID: 31853915. **Quality of life:** severely impacted by respiratory dependence (ventilation/tracheostomy), feeding difficulties, and intellectual disability requiring lifelong special-needs support.

4. Genetic / Molecular Information

Locus: 14q32.2 DLK1-DIO3 (DLK1-MEG3) imprinted cluster. **Paternally expressed:** DLK1 (HGNC:2907), RTL1/PEG11, DIO3 (HGNC:2885). **Maternally expressed (lost in KOS):** MEG3/GTL2 (HGNC:14575), RTL1as, MEG8, and clustered miRNAs (incl. miR-127 / miR-379–miR-544) and snoRNAs. **Variant classes:** structural (UPD, microdeletions) and epigenetic (DMR hypermethylation) — classic coding SNVs are *not* the usual mechanism. **Functional consequence:** gene dosage gain (overexpression) of paternal genes + loss of maternal ncRNA repression. **Epigenetics:** aberrant hypermethylation of the maternal IG-DMR (MEG3/DLK1:IG-DMR) and MEG3:TSS-DMR, the master and secondary control regions (PMID: 12937418, PMID: 35953028). **Chromosomal abnormalities:** Robertsonian translocations predisposing to upd(14)pat (PMID: 32211354).

5. Environmental Information

No environmental, lifestyle, or infectious contributors are established. KOS is a constitutional (epi)genetic disorder.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the full ordered causal chain. In brief: imprinting imbalance at 14q32.2 → loss of maternal ncRNAs (esp. RTL1as/miR-127) → RTL1 and DLK1 overdosage → endothelial/pericyte pathology (placentomegaly, polyhydramnios) + musculoskeletal dysplasia (coat-hanger ribs, small thorax) → restrictive respiratory failure, plus CNS (developmental delay) and hepatic (hepatoblastoma) branches. DLK1 acts in the Notch/Delta developmental-signaling axis; RTL1 is a retrotransposon-derived gene essential for fetal capillary maintenance.

7. Anatomical Structures Affected

- **Organ/system (primary):** thoracic skeleton/ribs (UBERON:0002228 rib; UBERON:0000915 thoracic segment), lungs (UBERON:0002048) — restrictive physiology; placenta (UBERON:0001987). **Secondary:** liver (UBERON:0002107, hepatoblastoma); CNS/brain (UBERON:0000955, developmental delay); abdominal wall (UBERON:0003697).
- **Tissue/cell:** placental **villous vascular endothelial cells (CL:0000115)** and **pericytes (CL:0000669)** (PMID: 22917972); skeletal muscle (CL:0000188); bone-forming cells.
- **Subcellular:** nucleus (GO:0005634) — site of imprinting control/methylation; plasma membrane (DLK1 is a transmembrane protein).

• **Lateralization:** bilateral/symmetric skeletal involvement.

8. Temporal Development

Onset: congenital; often detectable prenatally (polyhydramnios, small thorax, placentomegaly on ultrasound, PMID: 33179779). **Pattern:** chronic, non-episodic. **Course:** the critical period is the **neonatal interval**, when respiratory insufficiency determines survival; survivors have a static-to-slowly-evolving course with persistent developmental delay. Hepatoblastoma risk is concentrated in **infancy**. No spontaneous remission; lifelong condition.

9. Inheritance and Population

Prevalence/incidence: unknown/unrecorded — >80 cases reported in the literature; likely underdiagnosed (PMID: 34055463, PMID: 31853915). **Inheritance:** usually sporadic (de novo UPD or epimutation); microdeletion and translocation-associated cases can recur in families, warranting genetic counseling. **Penetrance:** developmental delay effectively 100% in classic molecular cases; **expressivity variable** (epimutation/mosaic cases milder, PMID: 35953028). **Founder effects / carrier frequency / ethnic predilection:** none established. **Sex ratio:** no strong skew documented (both male and female cases reported).

10. Diagnostics

Imaging: chest X-ray showing coat-hanger ribs + small bell-shaped thorax is the key diagnostic sign; prenatal ultrasound shows polyhydramnios, narrow thorax, protuberant abdomen, placentomegaly. **Confirmatory molecular testing (tiered):** MS-MLPA and methylation-specific PCR of IG-DMR/MEG3-DMR; SNP microarray; microsatellite/UPD analysis; karyotype/FISH (Robertsonian translocations); CNV-seq for microdeletions (PMID: 34746047, PMID: 36035167, PMID: 32211354). **Biomarker surveillance:** serum AFP for hepatoblastoma. **Differential diagnosis:** Beckwith-Wiedemann syndrome (overgrowth + hepatoblastoma), other skeletal dysplasias with small thorax, Pena-Shokeir phenotype; the coat-hanger rib sign distinguishes KOS.

11. Outcome / Prognosis

Overall prognosis is poor, driven by neonatal respiratory insufficiency (PMID: 34055463). Survivors have **universal developmental delay (median IQ/DQ ~55)** and require long-term ventilatory, feeding, and developmental support. **Hepatoblastoma (~8.8%)** is a key

complication requiring surveillance. Milder epimutation/mosaic cases can have near-normal development and better outcomes (PMID: 35953028). No validated prognostic biomarkers beyond molecular subtype (epimutation/mosaic → generally milder).

12. Treatment

No disease-modifying therapy exists. Management is entirely **supportive** (NCIT: Supportive Care): - **Respiratory:** mechanical ventilation, tracheostomy for chronic respiratory insufficiency. - **Surgical:** repair of abdominal wall defects (omphalocele/diastasis recti); hepatectomy if hepatoblastoma arises (PMID: 23181499). - **Nutrition/feeding support;** developmental/physical/occupational/speech therapy. - **Oncologic surveillance:** AFP + abdominal ultrasound.

No pharmacotherapy, gene therapy, or RNA-based therapy is approved. (Note: growth hormone benefits the *reciprocal* disorder, Temple syndrome, PMID: 30836360 — not applicable to KOS, where overgrowth is the phenotype.)

13. Prevention

No primary prevention exists (sporadic de novo origin in most cases). **Secondary prevention:** prenatal recognition of the ultrasound gestalt enabling molecular confirmation and informed pregnancy/postnatal planning (PMID: 33179779). **Tertiary prevention:** hepatoblastoma surveillance; respiratory support to prevent complications. **Genetic counseling** is indicated, especially for translocation- or deletion-associated familial recurrence risk.

14. Other Species / Natural Disease

The orthologous cluster is conserved in mouse (chr12qF1) and sheep. The **ovine callipyge** phenotype (*Ovis aries*, NCBI Taxon 9940) is a naturally occurring DLK1-DIO3 imprinting disorder of veterinary/agricultural relevance, producing muscular hypertrophy via cis up-regulation of DLK1/RTL1 (PMID: 24990181, PMID: 19816583). Orthologous genes: mouse *Rtl1/Peg11*, *Dlk1*, *Dio3*. The conservation of imprinting and dosage sensitivity across sheep, mouse, and human demonstrates deep evolutionary conservation of the mechanism.

15. Model Organisms

- **Mouse (NCBI Taxon 10090):** *Rtl1* knockout and overexpression (both lethal, PMID: 18176565); **miR-127 knockout** (placentomegaly, PMID: 26138477); **IG-DMR deletion** (imprinting control, PMID: 12937418); **uniparental disomy chr12** (muscle/bone/placenta

defects, PMID: 17449025); miR-379/miR-544 cluster knockout (muscle hypertrophy, PMID: 26487685). **Phenotype recapitulation:** strong for placentomegaly, dosage-lethality, and musculoskeletal effects. **Limitation:** no single mouse fully reproduces the complete human KOS gestalt (e.g., coat-hanger ribs).

- **Sheep:** callipyge as a natural dosage model (muscle-specific).

Evidence Base

PMID	Role in this report	Key supporting quote
22353941	Mechanism frequencies	"UPD(14)pat in 17 patients (65.4%), microdeletions ... 5 patients (19.2%), and epimutations in 4 patients (15.4%)"
26377239	Primary molecular cause	"excessive RTL1 expression and absent MEG expression constitute the primary underlying fact"
22917972	RTL1 dosage + placental cell pathology	"RTL1 expression level was about five times higher..."; "swollen vascular endothelial cells and hypertrophic pericytes"
25689926	Clinical gestalt, DD, hepatoblastoma	"developmental delay ... invariably present ... median ... quotient of 55"; "hepatoblastoma ... three infantile patients (8.8%)"; "+2.3 SD"
18176565	Rtl1 dosage-lethality	"both its loss and its overproduction cause late-fetal and/or neonatal lethality in mice"
26138477	miR-127 repressor	"miR-127 knockout mice exhibit placentomegaly with specific defects within the labyrinthine zone"
12937418	IG-DMR as master control	"deletion of the IG-DMR from the maternally inherited chromosome causes bidirectional loss of imprinting"
32592473	TS14/KOS mirror model	"imprinting disorders caused by absence or overexpression of genes within a single imprinted cluster on ... 14q32"
36577869	DLK1 dosage biology	"DLK1 is an imprinted gene, whose mutations have been described as a rare cause of CPP ... obesity and metabolic derangement"
34746047	Diagnostic approach	"Prenatal diagnosis using methylation-specific ... MLPA or a single-nucleotide polymorphism-based microarray ... is recommended"
24990181	Callipyge cis dosage	"A single nucleotide polymorphism in the DLK1-DIO3 imprinted gene cluster alters gene expression..."
19816583	DLK1/RTL1 drive muscle reprogramming	"muscle-specific expression patterns of most genes were similar to DLK1 ..."

Supporting case/series literature: PMID: 33179779, PMID: 34055463, PMID: 31853915, PMID: 32211354, PMID: 36035167, PMID: 23181499, PMID: 35953028, PMID: 25656753.

Limitations and Knowledge Gaps

1. **Epidemiology is unquantified.** True prevalence/incidence of KOS is unknown; the disorder is rare (>80 reported cases) and probably underdiagnosed. Sex ratio, geographic distribution, and carrier frequencies are not established.
2. **Relative contribution of RTL1 vs DLK1.** While RTL1 overdosage is identified as the primary driver, the precise, tissue-specific contribution of DLK1, DIO3, and lost maternal ncRNAs to each phenotype (ribs vs CNS vs liver) is not fully dissected in humans.
3. **Hepatoblastoma mechanism is inferred.** The ~8.8% risk is well documented, but the molecular link between 14q32.2 dosage and hepatoblastomagenesis is extrapolated from the shared imprinting-disorder biology with Beckwith-Wiedemann syndrome rather than directly demonstrated.
4. **Genotype–phenotype correlation is incomplete.** Epimutation/mosaic cases are milder, but predictors of severity (methylation level, tissue mosaicism distribution) are not standardized.
5. **No mouse model fully recapitulates the human gestalt**, limiting mechanistic study of specific features such as coat-hanger ribs.
6. **No therapeutics pipeline.** There are no disease-modifying treatments, and no clinical trials were identified; the imprinting nature of the disorder makes classical gene-replacement approaches conceptually difficult.
7. **This report is literature-derived.** No primary patient-level dataset was analyzed; findings rest on published cohorts and case reports (aggregated disease-level evidence).

Proposed Follow-up Experiments / Actions

1. **Establish a KOS registry** to quantify prevalence, incidence, survival, and genotype-stratified outcomes — directly addressing the epidemiology gap.

2. **Standardize methylation-based severity prediction:** correlate quantitative IG-DMR/MEG3-DMR methylation (and tissue mosaic ratio) with respiratory severity and DQ/IQ to build a prognostic model for epimutation cases (PMID: 35953028).
3. **Dissect RTL1 vs DLK1 contributions** using tissue-specific conditional mouse overexpression (rib/thorax, CNS, hepatocyte) to attribute individual phenotypes to specific gene overdosage.
4. **Interrogate the hepatoblastoma mechanism** via molecular profiling (transcriptomics/methylomics) of KOS-associated tumors vs sporadic hepatoblastoma to test the shared-imprinting hypothesis.
5. **Explore allele-specific epigenetic editing** (e.g., CRISPR-dCas9 demethylation targeting the hypermethylated maternal IG-DMR in epimutation cases) as a proof-of-concept corrective strategy in patient-derived iPSC/organoid models.
6. **Single-cell/spatial transcriptomics of KOS placenta and thoracic mesenchyme** to resolve the cell-type-specific dosage effects already localized to villous endothelium and pericytes (PMID: 22917972).
7. **Prospective hepatoblastoma surveillance protocol evaluation** (AFP + ultrasound interval optimization) to validate the recommended surveillance against outcomes.

Report compiled from 7 confirmed findings and 33 reviewed papers over 5 investigation iterations. Evidence type: predominantly human clinical cohort/case data, supported by mouse and ovine model-organism genetics and in vitro expression studies.
